Q7 $F_g - N = F_c$

$$\frac{GMm}{R^2} - N = mR\left(\frac{2\pi}{T}\right)^2$$

Since $N = 0$,

$$\frac{GMm}{R^2} = mR\left(\frac{2\pi}{T}\right)^2$$

$$T = 2\pi\sqrt{\frac{R^3}{GM}}$$

Answer: C

Q8 acceleration, $a = a_0 \sin\omega t$